

Asha Bhandarkar

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EDUCATION

Georgia Institute of Technology | Atlanta, GA

Master of Science in Electrical and Computer Engineering

August 2021 – December 2022

Bachelor of Science in Computer Engineering with Honors

August 2018 – May 2021

SKILLS

Languages: C, C++, Python, MATLAB

Embedded Systems: TDA4/AM62x SoCs, ESP32, AVR/Arduino, ARM Cortex-M, NVIDIA Jetson, Embedded Linux, FreeRTOS

Protocols: Bluetooth LE, I2C, SPI, UART, JTAG, CAN

Software & Debug: GDB, Trace32, Static Analysis, MISRA C/C++, Code Coverage (Klocworks, LDRA), Oscilloscope, Logic Analyzer

Development Tools: Git, Docker, Code Composer Studio, STM32CubeMX, PCB Design (EAGLE, KiCAD)

EXPERIENCE

Texas Instruments – Dallas, TX | Software Engineer

February 2023 – present

- Develop ASIL-B compliant AI algorithms for the C7x DSP + Matrix Multiplication Accelerator on TI Jacinto and Sitara SoCs
- Re-engineer the convolution operator for the TDA5 DSP architecture, achieving up to 3× speedup for depthwise convolutions while expanding supported model topologies and floating-point capabilities
- Collaborate with IP design, compiler, and deep-learning teams to verify IP, benchmark performance, debug simulator and compiler issues, and resolve software integration challenges
- Led DSP firmware, algorithms, and EdgeAI application development support for 10+ automotive Tier-1s/OEMs, resolving software defects and silicon errata to improve software quality and inform future SoC designs
- Optimized C7x math and vision algorithms on TDA4x and AM62x using specialized intrinsics and hardware features, creating the first open-source C7x algorithm examples and improving performance over the TI C66x DSP
- Reduced secondary bootloader runtime and footprint by developing ROM Combined Boot on ARM Cortex-R5F for TDA4 SoCs

Contextual Computing Group – Georgia Tech | Graduate Research Assistant

January 2022 – December 2022

- Researched wearable computing interfaces and passive haptic learning under the advisement of Dr. Thad Starner
- Designed and implemented hardware/firmware for Bluetooth LE-enabled, ESP32-based wireless haptic glove interface

Amazon Robotics – North Reading, MA | Software Development Engineer Intern

May 2022 – August 2022

- Designed and implemented a distributed computing solution for the Sparrow item-picking and manipulation robot
- Optimized segmentation and barcode decoding algorithms for speed and memory on NVIDIA Jetson AGX Orin platform

John Deere – Dubuque, IA | Embedded Systems Development Intern

May 2021 – August 2021

Intelligent Decision & Autonomous Robots Lab – Georgia Tech | Undergraduate Research Assistant

August 2020 – May 2021

Georgia Tech Research Institute – Atlanta, GA | Robotics Research Intern

May 2019 – August 2019

PROJECTS

TDA5 EdgeAI Performance Analysis and Optimization – Texas Instruments

April 2025 – December 2025

- Led cycle-accurate modeling project for the TDA5 AI subsystem in Synopsys Platform Architect for pre-silicon benchmarking
- Validated architectural accuracy and resolved integration issues with SystemC models for platform evaluation

RoboCup SSL, Intelligent Ground Vehicle Competition (IGVC) – RoboJackets, Georgia Tech

August 2018 – May 2022

- Designed and programmed PCB for robot identification and wrote firmware for STM32F7 chip
- Established robot's system design, including a custom CAN diagnostic system and PCB to increase range of E-stop system

MicroFloats: A Swarm of Underwater Autonomous Vehicles – Senior Design Project, Georgia Tech

January 2021 – May 2021

- Developed underwater vehicle electrical system including a buoyancy engine, GPS, LoRa, custom PCB, and sensor suite

LEADERSHIP

The Hive Makerspace – Georgia Tech | Volunteer Instructor

January 2019 – December 2022

RoboJackets – Georgia Tech | Electrical Lead

May 2019 – May 2021

- Lead a 15-member subteam to design a robot electrical system and coordinated with mechanical and software teams
- Managed team of 10+ members to create and present open-source electronics curriculum and circuit board to 40+ students

PUBLICATIONS & PRESENTATIONS

A.Bhandarkar. System-Level Benchmarking of AI Workloads for TDA5 Performance Analysis and Optimization. Synopsys Virtual Prototyping Day. Jun. 24, 2026.

A. Bhandarkar, T. Gemicioglu, B. Dwivedi, C. Seim, T. Starner. Learning Piano Songs with Passive Haptic Training: an Interactive Lesson. Proc. ACM Intl. Conf. Pervasive & Ubiquitous Comput., Sept. 2022. (Best Demo at ACM UbiComp/ISWC 2022 Conference)